RAM CIRIGIRI

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Engineering Leader, AI Systems

PROFESSIONAL SUMMARY

Engineering leader with 15+ years building high-scale distributed systems and production-grade AI platforms. Proven track record managing cross-functional teams (8+ engineers) at Fortune 25 companies delivering $5M+ in annual business impact. Specialist in real-time voice AI, multi-agent orchestration, and RAG pipelines — from research to production with strict quality, compliance, and cost-efficiency standards.

TECHNICAL CORE

Engineering Leadership	0→7 Team Building · Promotions · Technical Roadmap Ownership · Cross-functional Delivery · Engineering Standards
AI Architecture & Agents	LangGraph · Multi-agent Systems · RAG · Hybrid BM25+Vector · Prompt Engineering · Vector Search · Cosmos DB
AI Quality & Compliance	Eval Suites (CI-gated) · LLM Audit Logging · Hallucination Guard · PII Redaction · Feedback-to-Golden-Dataset
Voice & Real-time	Deepgram · ElevenLabs · Sarvam AI · Dual-STT Arbitration · WebSocket Streaming · Barge-In · High-Concurrency
Cloud & Infrastructure	Azure · Cosmos DB · Event Hubs · Python · FastAPI · C#.NET · Redis · Kafka · Kubernetes · Docker · Terraform

Technologies: LangGraph · Claude API · OpenAI API · Azure · Cosmos DB · Redis · Kafka · FastAPI · Deepgram · Sarvam AI · Docker · Kubernetes · Terraform · Next.js · WebSocket · RAG · BM25 · Vector Search

REPRESENTATIVE EXPERIENCE

Cognitive Commerce Labs (CCL) — June 2025 – Present
Founder — Austin, TX

Set the technical direction for two production-grade conversational AI systems, owning architecture end-to-end — independent research into multi-agent orchestration, real-time speech arbitration, and hybrid retrieval at scale.

- Invented a dual-engine speech-arbitration method (provisional patent, USPTO #64/026,827) that selects the more accurate transcript in ~1ms, improving recognition accuracy on accented and multilingual speech; built it into a real-time, three-channel conversational ordering agent (voice, chat, and phone).
- Engineered the agent as an 18-node LangGraph state machine with 32 grounded tools, completing most orders in a single exchange; a deterministic fast-path and tiered model routing (~70% of turns on a low-cost model) cut ~1s of latency per turn.
- Built an AI product-discovery engine architected for grocery catalogs of 2M+ SKUs, using a tiered resolution pipeline (deterministic matching → embeddings → LLM for only the ~3% long tail) and an ML entity-resolution flywheel reaching 95%+ automated resolution in testing.
- Achieved 85% Precision@10 (+15–20%) with hybrid BM25 + vector retrieval (RRF fusion) and cut P95 latency 800ms → 150ms via async execution and query optimization; load-tested to 100+ QPS at 99.9% availability.
- Held serving cost to roughly 1% of a naive LLM-per-query design through tiered routing and a two-tier cache (80%+ hit rate).
- Applied production-grade engineering throughout: speech- and LLM-provider failover, per-session distributed locking, and PII redaction on every model call; designed the canonical-normalization architecture to scale to 10,000 catalogs.
- Instrumented per-stage latency tracing across all 18 pipeline nodes and a structured LLM audit log (model, prompt version, tenant, token usage) to Azure Application Insights — with p95 (>2s) and error-rate (>1%) alerting, per-tenant cost attribution, and usage / cart-conversion dashboards, backed by 630 CI-gated regression tests guarding every release.

USPTO Provisional Patent (2026):

→ Domain-Aware Multilingual Speech Recognition Arbitration · #64/026,827

General Motors (Fortune 25) — 2022 – May 2025
Software Engineering Manager — Austin, TX

- **People Leadership** Grew team 0→7 (5 FTE + 2 offshore) for a greenfield claims platform targeting a $300B insurance market. Promoted Senior→Tech Lead, developed Mid→Senior through module ownership; navigated one PIP to resolution. Architectural patterns adopted as the global reference implementation. Engineers developed under this model have subsequently moved into senior technical leadership roles.
- **AI & Compliance** Led Claims Document Intelligence Tool — HIPAA-compliant architecture with PII/PHI redaction and customer-managed encryption keys, establishing the first safe GenAI use for sensitive insurance records at GM. Established the architectural and commercial foundation for $2M+ in projected annual savings — validated with business and compliance stakeholders.

Publix Super Markets (Fortune 100) — 2019 – 2022

Technical Programme Lead & Lead Architect — Lakeland, FL · 6-engineer cross-functional team · 2 concurrent programmes

- **Sales360** Architected a high-throughput event-driven platform (on-prem Kafka → Azure Event Hubs) as single source of truth for 1,200+ stores. Eliminated a 23-hour data staleness window via real-time streaming. Zero incidents at production cutover.
- Designed a two-tier caching architecture (in-memory L1 + Redis L2) for Avro schema resolution in an Azure Function processing 5M+ daily Kafka transactions at 500 TPS.
- **Central Refunds** Real-time compliance and fraud prevention platform · sub-6s SLA · configurable rules engine · full audit trail · $3M annual savings.

American Express (Fortune 100) — 2013 – 2019

Technical Lead — IVR & Customer Service Platform — Clearwater, FL · Team of 5 · Technology Champion

- Led Amex Serve platform processing $5.5M daily transactions (peak $20M). Designed intelligent IVR routing that reduced live agent transfers, saving $500K+ annually.

Inmarsat (Satellite Communications) — 2009 – 2013

Senior Software Engineer — Melbourne, FL

- Developed mission-critical satellite communications software for global maritime and aviation customers. Deep foundation in high-reliability distributed systems and fault-tolerant real-time data processing.

EDUCATION

MBA
University of Florida — Warrington College of Business

MS in Computer Science
Florida Institute of Technology — Distributed Systems & Software Engineering